BT503 – ENVIRONMENTAL BIOTECHNOLOGY

MID TERM IMPORTANT QUESTIONS

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Admin: **BS Biological Sciences** (One place to all your solutions)

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1. What are basic requirements of all cloning techniques?

Basic Requirements of Cloning Techniques

1. ***DNA insert:*** Target DNA to be cloned.
2. ***Cloning vector:*** Carries the target DNA into the host cell.
3. ***DNA-modifying enzymes:*** Restriction enzymes and DNA ligase, where required.
4. ***Host cell:*** Allows replication of recombinant DNA.
5. ***Selectable marker:*** Identifies cells containing the vector.
6. ***DNA delivery method:*** Introduces recombinant DNA into the host cell.
7. ***Screening and confirmation:*** Identifies and verifies the desired clone.

2. What compounds are used as bio filters?

Compounds Used as Biofilters

Common materials used as biofilters include:

1. Activated carbon
2. Peat
3. Compost
4. Soil
5. Wood chips and bark
6. Sand and gravel

3. Write steps of Southern blotting.

Steps of Southern Blotting

Southern blotting is a molecular biology technique used to detect a specific DNA sequence in a DNA sample.

1. DNA Extraction: DNA is isolated from the cells or tissue sample.

2. Restriction Digestion: The extracted DNA is cut into smaller fragments using restriction enzymes.

3. Gel Electrophoresis: DNA fragments are separated according to their size using agarose gel electrophoresis.

4. Denaturation: The double-stranded DNA fragments are treated with an alkaline solution to separate them into single strands.

5. Blotting (Transfer): The separated DNA fragments are transferred from the gel onto a nylon or nitrocellulose membrane.

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6. Fixation: DNA is attached firmly to the membrane using UV light or heat.
7. Hybridization: A labeled DNA probe complementary to the target sequence is added to the membrane. The probe binds to the specific DNA sequence.
8. Washing: The membrane is washed to remove unbound probes and reduce nonspecific binding.
9. Detection: The labeled probe is detected using an appropriate method, such as autoradiography or chemiluminescence, to identify the target DNA sequence.

4. What are maturation ponds?

Maturation Ponds

Maturation ponds are shallow ponds used in the final stage of wastewater treatment to remove harmful microorganisms, particularly pathogenic bacteria and viruses. They receive wastewater that has already undergone treatment in earlier ponds. The removal of pathogens occurs through the combined effects of sunlight, high pH, dissolved oxygen, temperature, and natural biological processes. Algae produce oxygen through photosynthesis, while sunlight and other environmental conditions help inactivate pathogens. Maturation ponds improve the quality of treated wastewater and reduce the risk of waterborne diseases. They also make the treated water more suitable for safe discharge into the environment or reuse for irrigation.

5. What are other ways to manipulate bacteria without genetic engineering?

Other Ways to Manipulate Bacteria Without Genetic Engineering

Bacteria can be manipulated in several ways without directly modifying their genetic material. These methods include:
1. Selective Culturing: Growing bacteria under specific conditions to encourage the growth of desired characteristics.
2. Environmental Modification: Changing temperature, pH, oxygen levels, or nutrient availability to influence bacterial growth and activity.
3. Adaptive Laboratory Evolution: Exposing bacteria to particular environmental conditions over many generations to select naturally occurring variants with useful traits.
4. Chemical Treatment: Using certain chemicals to alter bacterial growth, metabolism, or enzyme activity without intentionally changing their DNA.
5. Physical Treatment: Applying heat, cold, or radiation to influence bacterial survival and activity.
6. Co-culture: Growing two or more bacterial species together to influence their growth, metabolism, or interactions.

6. What methods are used in bio filters?

Methods Used in Biofilters

Biofilters use microorganisms to remove pollutants from air or water. The main methods include:
1. Biofiltration: Polluted air or water passes through a filter medium containing microorganisms that degrade contaminants.
2. Biosorption: Microorganisms or biological materials bind pollutants, such as heavy metals, to their surfaces.
3. Biodegradation: Microorganisms break down organic pollutants into simpler, less harmful substances.
4. Biooxidation: Microorganisms oxidize pollutants into less harmful products.
5. Biological Treatment: Microorganisms remove contaminants by using them as sources of energy or nutrients.

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7. What are five ways soil can be polluted due to pollutants?

Five Ways Soil Can Be Polluted

1. Industrial Waste: Discharge of industrial chemicals, heavy metals, and toxic substances contaminates the soil.

2. Agricultural Chemicals: Excessive use of pesticides, herbicides, and chemical fertilizers introduces harmful substances into the soil.

3. Improper Waste Disposal: Dumping household garbage, plastics, and other solid waste causes soil contamination.

4. Oil Spills and Leakage: Leakage of petroleum products from vehicles, pipelines, and storage tanks pollutes the soil.

5. Sewage and Wastewater: Untreated sewage and wastewater introduce pathogens, toxic chemicals, and other contaminants into the soil.

8. Write advantages of activated sludge in the process of wastewater treatment.

Advantages of Activated Sludge in Wastewater Treatment

The activated sludge process is a biological wastewater treatment method in which microorganisms break down organic pollutants in wastewater under aerobic conditions.

1. Removal of Organic Matter: Microorganisms decompose organic pollutants, reducing the biological oxygen demand (BOD) of wastewater.

2. High Treatment Efficiency: It effectively removes suspended solids and biodegradable organic substances.

3. Pathogen Reduction: The process helps reduce certain harmful microorganisms in wastewater.

4. Nutrient Removal: Under suitable operating conditions, it can help remove nitrogen and phosphorus compounds.

5. Less Land Requirement: It requires less land than many conventional pond-based treatment systems.

6. Sludge Recycling: Settled sludge can be returned to the aeration tank to maintain a sufficient population of microorganisms.

7. Improved Effluent Quality: It produces treated wastewater suitable for further treatment or discharge, subject to required environmental standards.

9. Write working of bio scrubbers.

Working of Bio Scrubbers

A bio scrubber is an air pollution control system that uses microorganisms to remove harmful gases and pollutants from contaminated air. The working of a bio scrubber involves the following steps:

1. Gas Absorption: Polluted air passes through a scrubber, where contaminants are absorbed into a liquid, usually water.

2. Transfer to Bioreactor: The pollutant-containing liquid is transferred to a biological reactor containing microorganisms.

3. Biodegradation: Microorganisms break down the absorbed pollutants into simpler, less harmful substances, such as carbon dioxide, water, and inorganic salts.

4. Liquid Recirculation: The treated liquid is often recirculated to the scrubber to improve efficiency and reduce water consumption.

5. Release of Clean Air: The purified air leaves the scrubber and is released into the atmosphere.

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10.What are advantages of using bio indicators?

Advantages of Using Bioindicators

Bioindicators are living organisms used to assess the quality of the environment and detect pollution or environmental changes.

1. ***Pollution Detection:*** They help detect pollutants in air, water, and soil.
2. ***Early Warning:*** They can indicate environmental problems before severe damage occurs.
3. Cost-Effective***:*** Monitoring bioindicators is often less expensive than repeated chemical analyses.
4. ***Assessment of Long-Term Effects:*** They can reveal the cumulative effects of pollution over time.
5. ***Ecosystem Health:*** They help assess biodiversity, ecosystem stability, and environmental quality.
6. ***Easy Monitoring:*** Some bioindicators, such as lichens and certain aquatic organisms, can be observed and monitored relatively easily.

11.Write difference between persistence and mobility.

Difference Between Persistence and Mobility

Persistence	Mobility
Persistence refers to how long a pollutant remains in the environment before it degrades or breaks down.	Mobility refers to the ability of a pollutant to move from one place to another in the environment.
Persistent pollutants remain in soil, water, or air for a long time.	Mobile pollutants can move through soil, water, or air.
It depends on the pollutant's chemical stability and environmental conditions.	It depends on solubility, soil properties, and environmental conditions.
Example: DDT persists in soil for a long time.	Example: Nitrate moves easily through soil and can contaminate groundwater.

12.Write name of three pathogens in environment.

Three common pathogens found in the environment are:

1. ***Bacteria:*** Escherichia coli (E. coli)
2. ***Viruses:*** Hepatitis A virus
3. ***Protozoa:*** Giardia lamblia

13.Enlist five criteria that classify a pollutant.

Five Criteria Used to Classify Pollutants

1. ***Origin:*** Natural or anthropogenic (human-made) pollutants.
2. ***Persistence:*** Biodegradable or non-biodegradable pollutants.
3. ***Nature:*** Physical, chemical, or biological pollutants.
4. ***Formation:*** Primary or secondary pollutants.
5. ***Medium affected:*** Air, water, or soil pollutants.

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14.Explain genetic modification of poplar plants. How they remove mercury from plants?

Genetic Modification of Poplar Plants for Mercury Removal

Genetically modified poplar plants are used in phytoremediation, a process in which plants remove or detoxify pollutants from contaminated soil and water. Scientists introduce specific genes into poplar plants to improve their ability to tolerate and detoxify mercury. Two important genes used in this process are:

1. ***merA gene:*** Obtained from mercury-resistant bacteria, it produces the enzyme mercuric reductase, which converts toxic ionic mercury (Hg^{2+}) into elemental mercury (Hg^{0}), which is more volatile and less toxic.
2. ***merB gene:*** It produces the enzyme organomercurial lyase, which breaks down certain organic mercury compounds into simpler compounds, allowing further detoxification by the MerA enzyme.

How mercury is removed:

- Poplar roots absorb mercury from contaminated soil or water.
- The introduced genes produce enzymes that convert mercury into less toxic forms.
- MerA converts ionic mercury into elemental mercury, which can escape into the atmosphere as a gas.
- The plants help reduce mercury contamination in the soil and water.

15.Enlist three main points regarding environmental biotechnology interventions.

Three Main Points Regarding Environmental Biotechnology Interventions

1. Pollution Control: Using microorganisms and plants to remove or degrade pollutants from air, water, and soil.

2. Waste Management: Applying biological processes to treat wastewater, decompose organic waste, and produce useful products such as biogas.

3. Environmental Restoration: Using bioremediation and phytoremediation to restore contaminated environments and improve ecosystem health.

16.Name three processes for removing pathogens from domestic sewage.

Three processes used to remove pathogens from domestic sewage are:

1. ***Chlorination:*** Chlorine is added to sewage to kill harmful microorganisms.
2. ***Ultraviolet (UV) Disinfection:*** UV radiation inactivates bacteria, viruses, and other pathogens.
3. ***Maturation Ponds:*** Sunlight, high pH, and natural biological processes help reduce pathogenic microorganisms.

17.Explain recombinant microorganisms particularly from bacteria and yeast.

Recombinant Microorganisms: Bacteria and Yeast

Recombinant microorganisms are bacteria or yeast whose genetic material has been modified by introducing DNA from another organism or by combining DNA sequences using genetic engineering techniques. They are used to produce useful proteins, medicines, enzymes, and other biological products.

1. Recombinant Bacteria

Bacteria, particularly Escherichia coli (E. coli), are widely used in genetic engineering. A desired gene is inserted into a plasmid vector and introduced into bacterial cells. The bacteria multiply and express the inserted gene to produce the required product. **Examples**: Recombinant E. coli is used to produce human insulin, growth hormone, and industrial enzymes.

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2. Recombinant Yeast

Yeast, particularly Saccharomyces cerevisiae, is a eukaryotic microorganism used to produce recombinant proteins. The desired gene is introduced into yeast cells using a suitable vector. Yeast can produce certain complex proteins and perform some post-translational modifications that bacteria cannot. **Examples**: Recombinant yeast is used to produce hepatitis B vaccine antigens, enzymes, and other therapeutic proteins.

18. Why maturation of pond is done after the preliminary treatment?

Maturation ponds are used in the later stages of wastewater treatment to reduce pathogenic microorganisms and improve the quality of treated sewage. However, they do not usually follow preliminary treatment immediately; they generally follow secondary treatment or other stabilization ponds. The main purposes of maturation ponds are:

1. ***Pathogen Removal:*** Sunlight, high pH, and other environmental conditions help destroy harmful bacteria, viruses, and parasites.
2. ***Water Quality Improvement:*** They help reduce remaining organic matter and suspended solids.
3. ***Public Health Protection:*** They reduce the risk of waterborne diseases.
4. ***Safe Disposal and Reuse:*** They improve wastewater quality for discharge into the environment or reuse for irrigation.

19. Genetic modification of poplar plant help in removal of mercury from soil. How it is done?

Genetic Modification of Poplar Plants for Mercury Removal

Genetically modified poplar plants are used in phytoremediation to reduce mercury contamination in soil. Scientists introduce genes from mercury-resistant bacteria into poplar plants to help them detoxify mercury. The process occurs as follows:

1. Gene Introduction: The merA gene from mercury-resistant bacteria is introduced into poplar plant cells using genetic engineering techniques.
2. Enzyme Production: The introduced gene produces the enzyme mercuric reductase inside the plant cells.
3. Mercury Absorption: The roots absorb mercury ions (Hg^{2+}) from contaminated soil along with water and minerals.
4. Mercury Conversion: Mercuric reductase converts toxic ionic mercury (Hg^{2+}) into elemental mercury (Hg^{0}), which is more volatile and less toxic.
5. Mercury Release: Elemental mercury can evaporate from the plant into the atmosphere, reducing the amount of mercury remaining in the soil.

20. What three attributes aqueous invertebrate must have for their role as a bio indicator?

Three Attributes of Aquatic Invertebrates as Bioindicators

1. Sensitivity to Pollution: They should respond predictably to changes in water quality and pollution levels.
2. Wide Distribution: They should be present in many aquatic habitats, allowing comparison of water quality between different locations.
3. Limited Mobility: They should not move extensively, so their presence or absence reflects the environmental conditions of a particular site.

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21.What are thermophiles and their location?

Thermophiles are microorganisms that grow optimally at high temperatures, usually between 45°C and 80°C. Some extreme thermophiles can grow at even higher temperatures. Locations of thermophiles:

1. ***Hot springs:*** Found in naturally heated water sources.
2. ***Hydrothermal vents:*** Live in hot environments on the ocean floor.
3. ***Geothermal areas:*** Found in hot soils and volcanic regions.
4. ***Compost heaps:*** Occur in decomposing organic matter where temperatures become high.

22.Briefly describe cDNA.

Complementary DNA (cDNA)

Complementary DNA (cDNA) is a DNA molecule synthesized from a messenger RNA (mRNA) template with the help of the enzyme reverse transcriptase. This process is called reverse transcription. In eukaryotic cells, cDNA is usually prepared from mature mRNA, so it generally lacks introns and contains the sequences of expressed genes. The synthesized cDNA can be amplified using PCR or inserted into a cloning vector for further study.

Applications of cDNA:

1. Used to study gene expression.
2. Helps in gene cloning and recombinant protein production.
3. Used in RT-PCR to detect specific RNA transcripts.
4. Helps in preparing cDNA libraries for research.

23.Define denitrification. Name two denitrifying agents.

Denitrification: Denitrification is the biological process in which bacteria convert nitrates into gaseous nitrogen under anaerobic or low-oxygen conditions. Two denitrifying agents are:

1. *Pseudomonas denitrificans*
2. *Paracoccus denitrificans*

24.Define transduction, transformation, plasmid and conjugation.

1. Transduction: Transduction is the transfer of genetic material from one bacterium to another through a bacteriophage (bacterial virus).

2. Transformation: Transformation is the process by which a bacterium takes up free DNA from its surrounding environment and incorporates it into its genetic material.

3. Plasmid: A plasmid is a small, usually circular, double-stranded DNA molecule that replicates independently of the bacterial chromosome. It may carry genes for antibiotic resistance or other useful traits.

4. Conjugation: Conjugation is the transfer of genetic material from one bacterium to another through direct cell-to-cell contact, often involving a conjugative plasmid and a sex pilus.

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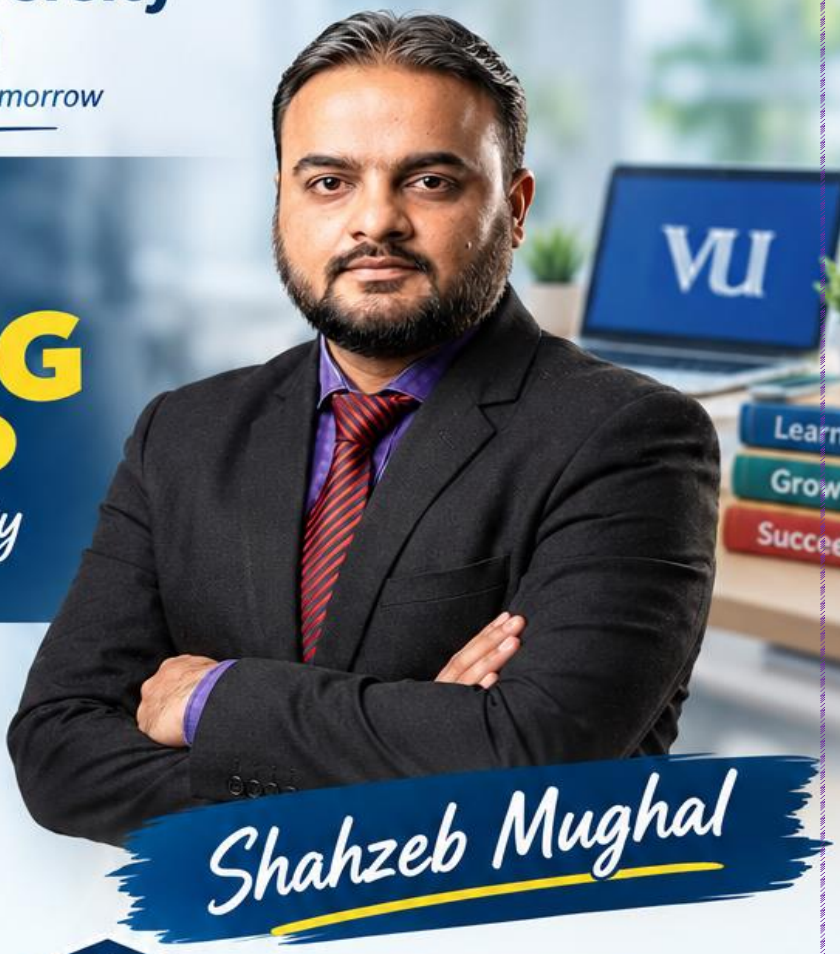
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